

EXPLAINABLE VISION-LANGUAGE DIAGNOSTIC ASSISTANT

for Chest Radiographs

Project Proposal & Literature Survey Report

Abstract

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Guide	Dr. K. Swanthana
Domain	Medical AI / Computer Vision / Vision-Language Models
Deployment Goal	Offline, edge-optimized assistant for rural healthcare settings

1. Project Title

Explainable Vision-Language Diagnostic Assistant for Chest Radiographs

An offline, edge-optimized medical AI solution that combines automated chest X-ray interpretation, visual explanation through heatmaps and attention overlays, plain-language diagnostic descriptions, and priority triage guidance for use by rural health workers.

2. Problem Statement

Primary health centres in rural regions may face limited access to specialist radiologists. This can delay interpretation of chest radiographs and increase the need for patients to travel to urban hospitals. Existing AI systems can provide disease classifications, but a prediction without a clear visual or textual explanation can be difficult for non-specialist healthcare workers to trust and interpret. The proposed project addresses this gap by building an explainable multimodal assistant that accepts a chest X-ray, identifies likely thoracic abnormalities, highlights relevant image regions, generates a simple diagnostic description, and assigns a priority/urgency flag.

- Acute radiologist shortage: rural health centres may not have immediate specialist interpretation.
- Diagnostic delays and travel burden: patients may need to travel to urban hospitals for specialist review.
- Black-box AI mistrust: class probabilities alone do not show why a model reached its conclusion.
- Connectivity constraints: cloud-only AI is unsuitable for locations with unreliable or absent internet.

3. Objectives

Multimodal Diagnosis: Develop a vision-language assistant that produces disease predictions/confidence scores together with plain-language diagnostic descriptions such as possible early signs of pneumonia.

Visual Explanation: Generate Grad-CAM heatmaps and attention overlays that identify image regions contributing to the prediction.

Automated Triage & Priority Guidance: Convert model outputs into a conservative urgency flag, for example, recommending specialist review within a defined time window for higher-priority cases. This is decision support, not a replacement for a clinician.

Offline Edge Deployment: Use model optimization and INT8 quantization to target execution on a mid-range Android device without requiring continuous internet connectivity.

Evaluation: Measure classification quality, explanation/localization quality, text-generation quality, and mobile runtime performance.

4. Proposed Model / Framework

The proposed framework is a dual-stream vision-language architecture inspired by the components presented in the supplied project PPT. The PPT proposes combining Vision Transformers (ViT) for spatial feature extraction, CLIP for vision-text alignment, and BLIP-2 for diagnostic text generation, with Grad-CAM and attention overlays for explanation and INT8 quantization for offline Android deployment.

4.1 Proposed Architecture

- 1. Input Layer: chest radiograph (PNG/JPEG/DICOM converted to the required model input format).
- 2. Pre-processing Layer: resizing, normalization, quality checks and medically appropriate augmentation during training.
- 3. Vision Encoder: ViT-based image feature extraction for spatial representations.
- 4. Vision-Text Alignment: CLIP-style/medical contrastive alignment to map image and text representations into a shared space.
- 5. Disease Prediction Head: multi-label thoracic disease classification with calibrated confidence scores.
- 6. Language Generation Head: BLIP-2-inspired image-to-text generation for a concise, plain-language diagnostic description.
- 7. Explainability Layer: Grad-CAM heatmap for class-specific localization plus cross-attention visualization where supported.
- 8. Triage Layer: rule-based conservative priority mapping from validated model outputs; examples include routine review, priority review, or urgent specialist review.
- 9. Optimization Layer: pruning/precision reduction as appropriate, followed by INT8 quantization and ONNX/mobile runtime conversion.
- 10. Application Layer: Android interface displaying X-ray, predictions, confidence, explanation heatmap, generated description and triage flag.

4.2 Why This Framework Addresses the Limitations

Limitation	Proposed Response
Classification-only systems	Add language generation and multimodal reasoning rather than returning only labels.
Poor transparency	Add Grad-CAM and attention overlays tied to the predicted abnormality.
Paired-data dependency	Use both paired image-report datasets and image-only datasets where compatible with the training stage.
Heavy cloud inference	Optimize and quantize the final model for offline edge execution.
Lack of actionable prioritization	Add a clearly separated triage-support module based on validated thresholds and safety rules.

5. Literature Survey

The following survey expands the literature direction shown in the supplied PPT. The first five works directly correspond to major components named in the PPT (ResNet/DenseNet, CheXNet, Grad-CAM, BLIP-2 and MedKLIP), while additional medical vision-language and chest-X-ray works are included to reach the requested minimum of ten papers.

N o .	Research Paper Title	Model / Framework	Outcome	Limitations / Research Gap	Datasets Used	Paper Link
1	Deep Residual Learning for Image Recognition	ResNet / residual blocks	Introduced residual learning with shortcut	General computer-vision model; not designed for multimodal radiology,	ImageNet, CIFAR-10 and other recognition benchmarks in the original	https://arxiv.org/abs/1512.03385

No.	Research Paper Title	Model / Framework	Outcome	Limitations / Research Gap	Datasets Used	Paper Link
	tion		connections, enabling much deeper networks and strong image-recognition performance.	textual report generation, explainability or triage.	study.	
2	Densely Connected Convolutional Networks	DenseNet	Dense connections improve feature propagation and feature reuse while reducing parameter requirements relative to comparable deep networks.	Primarily an image-recognition architecture; does not inherently provide language reasoning, report generation or clinical triage.	CIFAR-10, CIFAR-100, SVHN and ImageNet.	https://arxiv.org/abs/1608.06993
3	CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with	121-layer DenseNet	Detected pneumonia from chest radiographs and extended the model	Primarily a classification /localization system; it does not provide a complete vision-language diagnostic conversation, natural-	ChestX-ray14; a separately annotated radiologist test set of 420 frontal chest X-rays was used for comparison.	https://arxiv.org/abs/1711.05225

No.	Research Paper Title	Model / Framework	Outcome	Limitations / Research Gap	Datasets Used	Paper Link
	Deep Learning		to all 14 ChestX-ray14 disease labels; reported performance exceeding the average radiologist on the study's F1 comparison.	language explanation or offline triage workflow.		
4	Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization	Grad-CAM	Produces class-discriminative heatmaps by using gradients to identify image regions important for a prediction.	Provides visual attribution but does not by itself generate a clinical textual explanation, severity assessment or triage recommendation.	Evaluated across image classification, captioning and VQA examples using multiple datasets/models.	https://arxiv.org/abs/1610.02391
5	BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language	BLIP-2 with Q-Former	Bridges frozen image encoders and frozen large language models using a lightweight Queryin	Large language model/image encoder components can still be computationally demanding for small offline devices; not specifically designed for	Multiple vision-language benchmarks including VQAv2 and image-text generation/evaluation tasks.	https://arxiv.org/abs/2301.12597

No.	Research Paper Title	Model / Framework	Outcome	Limitations / Research Gap	Datasets Used	Paper Link
	ee Models		g Transformer, achieving strong vision-language performance with fewer trainable parameters.	chest radiograph safety or triage.		
6	MedKLIP: Medical Knowledge Enhanced Language-Image Pre-Training in Radiology	MedKLIP	Uses medical knowledge, extracted medical entities and spatial alignment to strengthen radiology image-language pre-training ; evaluated on multiple disease - classification and grounding bench	More complex and computationally demanding than lightweight edge-oriented architectures ; it does not directly solve offline mobile deployment and triage.	ChestX-ray14, RSNA Pneumonia , SIIM-ACR Pneumothorax, COVIDx CXR-2, COVID Rural and EdemaSeverity.	https://arxiv.org/abs/2301.02228

No.	Research Paper Title	Model / Framework	Outcome	Limitations / Research Gap	Datasets Used	Paper Link
			marks.			
7	GLoRIA: A Multimodal Global-Local Representation Learning Framework for Label-Efficient Medical Image Recognition	GLoRIA	Learns global and local image-text representations by associating image sub-regions with report words; supports retrieval, classification and segmentation with improved label efficiency.	Requires paired image-report data for its multimodal training setup and is focused on representation learning rather than a complete diagnosis-to-triage assistant.	Medical imaging datasets including MIMIC-CXR and CheXpert in downstream experiments.	https://openaccess.thecvf.com/content/ICCV2021/html/Huang_GLoRIA_A_Multimodal_Global-Local_Representation_Learning_Framework_for_Label-Efficient_Medical_Image_Recognition_ICCV_2021_paper.html
8	Contrastive Learning of Medical Visual Representations from Paired Images and Text	ConVIRT	Uses bidirectional contrastive learning between paired medical images and reports; achieved strong transfer	Depends on paired image-text data and primarily learns transferable representations rather than directly generating explanations and actionable triage.	Four medical image classification tasks and two zero-shot retrieval tasks; radiology image-report data were central to pretraining.	https://arxiv.org/abs/2010.00747

No .	Research Paper Title	Model / Framework	Outcome	Limitations / Research Gap	Datasets Used	Paper Link
			performance and data efficiency on multiple medical image classification tasks.			
9	MedCLIP: Contrastive Learning from Unpaired Medical Images and Text	MedCLIP	Decouples images and texts and uses a medical - knowledge-based semantic matching loss, reducing false negatives and improving zero-shot prediction, supervised classification and image-text retrieval.	Primarily a representation/contrastive learning framework; it does not itself provide complete report generation, explanation heatmaps and edge triage.	Uses medical image/text resources including MIMIC-CXR, CheXpert and related medical datasets in evaluation /pretraining experiments.	https://aclanthology.org/2022.emnlp-main.256/
10	ChestX-ray8:	Weakly-supervised	Introduced a	Labels are text-	ChestX-ray8:	https://openaccess.thecvf.com/content_cvpr_2017/html/Wang_ChestX-ray8_Hospital-

No.	Research Paper Title	Model / Framework	Outcome	Limitations / Research Gap	Datasets Used	Paper Link
	Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases	multi-label CNN classification/localization	large chest X-ray benchmark and demonstrated weakly supervised detection and spatial localization of common thoracic diseases from image-level labels.	mined/weakly supervised; the benchmark is not a complete multimodal reasoning, report-generation or triage system.	108,948 frontal-view X-rays from 32,717 patients, with eight text-mined disease labels.	Scale_Chest_CVPR_2017_paper.html
11	Automated abnormality classification of chest radiographs using deep convolutional neural networks	Deep CNNs including AlexNet, VGGNet, ResNet, Inception-v3 and DenseNet	Compared deep CNN architectures for chest-radiograph abnormality classification and found strong performance, with transfer learning improving	Focuses on abnormality classification and model comparison rather than integrated vision-language explanations, triage and offline deployment.	NIH ChestX-ray14, Indiana-related chest radiographs and Guangzhou pediatric chest radiographs.	https://pmc.ncbi.nlm.nih.gov/articles/PMC7224391/

No.	Research Paper Title	Model / Framework	Outcome	Limitations / Research Gap	Datasets Used	Paper Link
			results.			

6. Datasets and Dataset Access Links

Dataset	Purpose / Key Details	Access Link
ChestX-ray14 / NIH ChestX-ray	112,120 frontal-view chest X-rays and 14 thoracic disease labels; the PPT identifies it as the primary large-scale classification source.	https://nihcc.app.box.com/v/ChestXray-NIHCC
OpenI / Indiana University Chest X-ray Collection	7,470 X-rays paired with clinical radiology reports; useful for image-text alignment and report-generation experiments.	https://openi.nlm.nih.gov/
ROCO Medical	Radiology image-text dataset suitable for multimodal representation and image-caption alignment experiments.	https://github.com/razorx89/ROCO
CheXpert	Large chest radiograph dataset with uncertainty labels and radiology reports; useful as an external evaluation/generalization dataset.	https://stanfordmlgroup.github.io/competitions/chexpert/
MIMIC-CXR	Large chest radiograph dataset with associated free-text reports; access requires registration and credentialing/data-use requirements.	https://physionet.org/content/mimic-cxr/2.0.0/

7. Research Gap Identification

Gap 1 — Classification without integrated reasoning

Many chest-X-ray systems focus on classification. The proposed project combines disease prediction with a plain-language diagnostic description so that the output is more understandable to the intended user.

Gap 2 — Explanation disconnected from diagnosis text

Grad-CAM and saliency methods can show relevant regions, but the visual explanation is often presented separately from a natural-language explanation. The proposed system explicitly links the predicted abnormality, confidence and visual evidence.

Gap 3 — Limited actionable triage

The surveyed multimodal representation models do not directly provide a practical urgency/priority layer. The project adds a conservative, rule-based triage-support module with clearly separated safety messaging.

Gap 4 — Heavy multimodal models versus rural connectivity

Vision-language models can be computationally expensive. The project explicitly targets quantization and offline Android execution rather than assuming continuous cloud access.

Gap 5 — Data limitations

Paired image-report datasets are smaller and harder to obtain than image-only datasets. The project therefore uses multiple public sources and separates pretraining/alignment data from supervised disease-classification data where appropriate.

Gap 6 — Evaluation beyond accuracy

A useful diagnostic assistant must be evaluated not only by accuracy but also by precision, recall, F1, localization/explanation quality and edge runtime performance.

7.1 How the Project Will Address the Gaps

- Build a modular multimodal pipeline instead of a single classifier.
- Generate both visual evidence and textual explanation for each prediction.
- Keep triage logic separate from generative text so safety-critical priority decisions are deterministic and auditable.
- Use quantization/ONNX or an equivalent mobile runtime to reduce memory and inference cost.
- Use patient-wise train/validation/test splitting to reduce leakage and evaluate generalization.
- Compare the proposed system against classification-only and explanation-only baselines.

8. Innovation, Creativity and Novelty

Aspect	Project Contribution
Innovation	Integration of vision-language diagnosis, visual explanation and offline edge deployment in one chest-radiograph assistant.
Creativity	The output is designed around the rural health worker's workflow: upload X-ray → inspect prediction → inspect evidence → read simple explanation → see priority guidance.
Novelty	The project focuses on the combination of multimodal interpretation + explainability + conservative triage support + offline Android execution rather than presenting a new foundation model from scratch.
Practical novelty	The final prototype is intended to remain useful under zero-connectivity conditions, which is a key deployment constraint in the stated problem context.

9. Feasibility

Area	Feasibility Assessment
Technical feasibility	The project can be implemented using established open-source architectures and pretrained components. The

	system should be modular so that a smaller vision model can be used if the complete BLIP-2/VLM stack is too heavy for the target device.
Dataset feasibility	ChestX-ray14, OpenI and ROCO are identified in the supplied PPT, and additional public datasets such as CheXpert can support validation/generalization.
Implementation feasibility	A staged implementation allows the team to first build classification, then explanation, then language generation, then quantization/mobile deployment.
Compute feasibility	Training/fine-tuning may require a GPU, but the final deployment target is a quantized inference model. If full multimodal training is too expensive, pretrained encoders can be frozen and only lightweight adapters/heads can be trained.
Deployment feasibility	The PPT specifically proposes INT8 quantization and ONNX/Android runtime integration. A fallback desktop/offline prototype can be maintained if full mobile optimization is not completed.
Safety feasibility	The prototype should be positioned as decision support and screening/triage assistance, not an autonomous medical diagnosis. Outputs should explicitly recommend professional review and expose uncertainty.

10. Complete Project Workflow / Execution Plan

The workflow below follows the four phases shown in the PPT and expands them into executable project tasks.

PHASE 1 — Data Preparation & Alignment

11. Collect and organize ChestX-ray14, OpenI and ROCO data; add CheXpert/MIMIC-CXR only if access and project scope permit.
12. Remove corrupt/duplicate files and standardize image formats.
13. Perform patient-wise train/validation/test splitting wherever patient identifiers are available.
14. Resize and normalize X-rays; apply medically appropriate augmentation to the training split only.
15. Prepare image labels for multi-label disease classification.
16. Prepare image-report pairs for vision-language alignment and report-generation experiments.
17. Create a consistent vocabulary/target format for plain-language output.

PHASE 2 — Model, Multimodal Alignment & Explainability

18. Establish a baseline CNN/DenseNet classifier.
19. Train or fine-tune the disease classification head.
20. Integrate ViT or another selected vision encoder for spatial features.
21. Add CLIP/medical contrastive alignment between image and text representations.
22. Integrate a BLIP-2-inspired generation component or a smaller image-to-text model if edge constraints require it.
23. Generate class-specific Grad-CAM heatmaps.
24. Generate cross-attention overlays when supported by the selected VLM.
25. Build the combined output: disease probabilities + explanation image + short text.

PHASE 3 — Optimization & Mobile Deployment

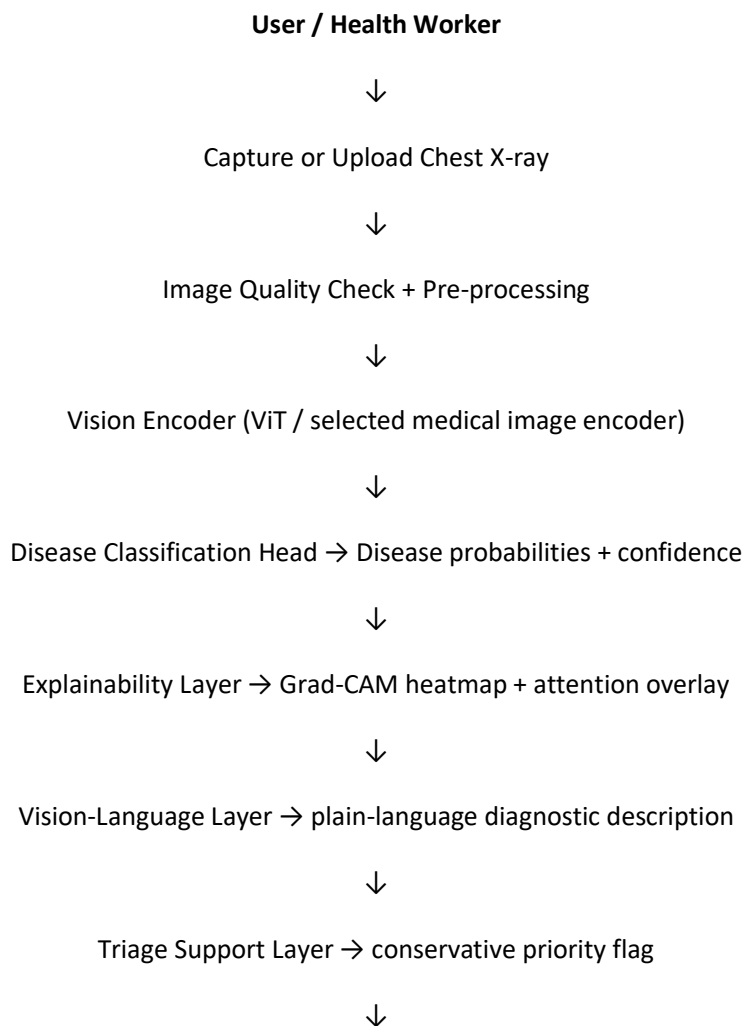
26. Profile model size, RAM use and inference latency.

27. Freeze large pretrained components where possible and reduce trainable parameters.
28. Apply FP16/INT8 quantization and verify that predictive performance remains acceptable.
29. Export compatible components to ONNX or an Android-supported inference runtime.
30. Build an Android interface for X-ray input and results.
31. Implement local storage and offline inference without requiring an API call.
32. Add safety messages and clear separation between model prediction and triage-support rules.

PHASE 4 — Evaluation, Validation & Presentation

33. Evaluate classification using Accuracy where appropriate, Precision, Recall, F1 and ROC-AUC.
34. Evaluate explanation quality using localization checks where ground-truth regions are available and/or expert qualitative review.
35. Evaluate generated text using task-appropriate language metrics plus factual consistency review.
36. Measure mobile latency, memory footprint and model size.
37. Run ablation studies: classifier only vs classifier + Grad-CAM; image-only vs image-text alignment; full precision vs INT8.
38. Test representative normal and abnormal cases and document failure cases.
39. Prepare final demo, report, architecture diagram, evaluation tables and presentation.

11. End-to-End System Flow



Offline Android UI → X-ray + prediction + visual evidence + text + safety guidance



Store/Export Result for clinician review

12. Evaluation Strategy

Component	Metrics	Evaluation Goal
Disease classification	Precision, Recall, F1-score, ROC-AUC, sensitivity/specificity where applicable	Compare with DenseNet/CheXNet-style baseline.
Visual explanation	Localization overlap/pointing-game style checks where annotations exist; qualitative expert review	Check whether highlighted regions correspond to plausible pathology regions.
Text generation	Clinical factuality review plus standard text similarity metrics where suitable	Penalize unsupported findings and hallucinated diagnoses.
Triage	Sensitivity for high-priority cases; false-negative analysis; rule consistency	Triage should be conservative and never presented as a definitive diagnosis.
Edge deployment	Latency, RAM, model size, battery/runtime behavior	Compare FP32/FP16/INT8 versions.
Robustness	Performance across datasets/subgroups where metadata permits	Investigate distribution shift and failure cases.

13. Proposed Project Plan / Timeline

Period	Work Package
Week 1	Dataset acquisition, cleaning, patient-wise split, preprocessing pipeline
Week 2	Baseline DenseNet/ResNet classifier and evaluation
Week 3	ViT/CLIP-style multimodal alignment experiments
Week 4	BLIP-2-inspired report generation / lightweight alternative
Week 5	Grad-CAM and attention visualization integration
Week 6	Triage-support logic, safety layer and integrated backend
Week 7	INT8 quantization, ONNX/mobile runtime optimization
Week 8	Android UI, offline testing and performance profiling
Week 9	Ablation studies, robustness testing, error analysis
Week 10	Final evaluation, documentation, demo and presentation

14. Expected Final Deliverable

- A working offline/edge prototype for chest X-ray decision support.
- Multi-label disease predictions with confidence scores.
- Visual Grad-CAM/attention explanations.
- Plain-language diagnostic descriptions.
- A conservative triage/priority indicator with safety disclaimer.

- Android/mobile demonstration or an equivalent offline prototype if device constraints require it.
- Experimental comparison against baseline models.
- Complete project report containing methodology, literature survey, research gap, experiments, results and limitations.

15. Anticipated Limitations and Risk Mitigation

Risk / Limitation	Mitigation
Dataset label noise	ChestX-ray14 labels are largely derived from reports; use patient-wise splits, external validation and error analysis.
Model hallucination	Constrain generated text to detected findings/templates where possible and add factual consistency checks.
Explanation faithfulness	Treat heatmaps as model-attribution evidence, not proof of pathology; validate localization and present uncertainty.
Mobile compute limits	Use frozen encoders, smaller models, quantization and staged deployment.
Distribution shift	Evaluate on an external dataset when possible and report limitations instead of claiming universal performance.
Clinical safety	Do not position the prototype as an autonomous diagnostic device; require clinician confirmation for medical decisions.

16. Key References and Links

BLIP-2: <https://arxiv.org/abs/2301.12597>

MedKLIP: <https://arxiv.org/abs/2301.02228>

Grad-CAM: <https://arxiv.org/abs/1610.02391>

CheXNet: <https://arxiv.org/abs/1711.05225>

GLoRIA: https://openaccess.thecvf.com/content/ICCV2021/html/Huang_GLoRIA_A_Multimodal_Global-Local_Representation_Learning_Framework_for_Label-Efficient_Medical_Image_Recognition_ICCV_2021_paper.html

ConVIRT: <https://arxiv.org/abs/2010.00747>

MedCLIP: <https://aclanthology.org/2022.emnlp-main.256/>

ChestX-ray8: https://openaccess.thecvf.com/content_cvpr_2017/html/Wang_ChestX-ray8_Hospital-Scale_Chest_CVPR_2017_paper.html

DenseNet: <https://arxiv.org/abs/1608.06993>

ResNet: <https://arxiv.org/abs/1512.03385>

NIH ChestX-ray14: <https://nihcc.app.box.com/v/ChestXray-NIHCC>

OpenI: <https://openi.nlm.nih.gov/>

ROCO: <https://github.com/razorx89/ROCO>

CheXpert: <https://stanfordmlgroup.github.io/competitions/chexpert/>

MIMIC-CXR: <https://physionet.org/content/mimic-cxr/2.0.0/>